

Sample Learning Notes

Official Statistics - Sampling and Survey Design

Prepared as a test document for Skillify's local Qwen/Ollama MCQ generation pipeline.

1. Introduction to Official Statistical Surveys

Official statistical surveys are systematic data-collection activities conducted or sponsored by public authorities to measure social, economic, demographic, agricultural, labour, and other characteristics of a population. A well-designed survey aims to produce estimates that are relevant, timely, accurate, and sufficiently precise for decision-making.

A typical survey cycle includes defining objectives, identifying the target population, designing the sample, preparing questionnaires and field procedures, collecting data, validating and processing observations, estimating population characteristics, and disseminating results.

2. Census and Sample Survey

Method	Meaning	Main Advantage	Main Limitation
Census	Data are collected from every unit in the target population.	Provides complete population coverage.	Usually expensive, time-consuming, and operationally demanding.
Sample Survey	Data are collected from a scientifically selected subset of the population.	Lower cost and faster data collection.	Estimates contain sampling error.

3. Target Population, Sampling Frame, and Sampling Unit

Target population: the complete set of units about which the survey seeks information. For example, all manufacturing establishments in a specified geographic area.

Sampling frame: a practical list or representation of population units from which the sample is selected. A frame should be as complete, accurate, and up-to-date as possible.

Sampling unit: the unit that is selected during a particular stage of sampling, such as a household, person, village, enterprise, or geographic area.

4. Simple Random Sampling

In simple random sampling, every eligible population unit has a known and equal probability of selection, and selection is carried out using a random mechanism. Simple random sampling is conceptually straightforward but may require a complete sampling frame and can become inefficient when the population is geographically dispersed.

For a population of size N and a sample of size n , the selection probability for each unit under equal-probability simple random sampling is n/N .

5. Stratified Sampling

Stratified sampling divides the population into non-overlapping, collectively exhaustive groups called strata, and then selects a sample from each stratum. A stratum is often defined using characteristics related to the study variable, such as geography, sector, size class, or demographic group.

A major benefit is that stratification can improve precision when units within a stratum are relatively homogeneous and the strata differ meaningfully from one another. It can also ensure that important

population groups are represented.

6. Systematic Sampling

In systematic sampling, units are arranged in an ordered frame, a sampling interval is calculated, and units are selected at regular intervals after a random starting point. If the population size is N and the desired sample size is n , a commonly used interval is approximately $k = N/n$.

Systematic sampling is easy to implement, but the ordering of the frame should be examined carefully. A periodic pattern that coincides with the sampling interval can introduce bias.

7. Cluster Sampling

Cluster sampling divides the population into groups called clusters, such as villages, schools, or geographic areas. Instead of selecting individual population elements directly, the design selects clusters, sometimes followed by one or more additional stages of sampling within the selected clusters.

Cluster sampling can reduce field travel and operational costs when population units are geographically dispersed. However, units within a cluster may be more similar to one another than units from different clusters, which can increase sampling variance compared with an equally sized simple random sample.

8. Sampling Error and Non-Sampling Error

Type	Description	Example
Sampling error	Difference between a sample-based estimate and the corresponding population value caused by using a sample instead of a census.	An estimate of average income differs from the true population average.
Coverage error	Error caused when the sampling frame does not properly represent the target population.	A list excludes newly created establishments.
Non-response error	Bias or loss of information because selected units do not provide usable responses.	Selected households repeatedly cannot be contacted.
Measurement error	Error caused by inaccurate responses, poor questions, interviewer effects, or recording problems.	A respondent misunderstands a question about employment status.
Processing error	Errors introduced during coding, data entry, editing, imputation, or transformation.	A valid value is incorrectly coded during data entry.

9. Non-Response and Response Rate

Non-response occurs when selected units do not provide the requested information. High or systematic non-response can reduce the quality of survey estimates. Response rate is commonly expressed as the proportion of eligible sampled units from which usable responses were obtained, subject to the definition used by the survey.

Strategies for reducing non-response include clear communication, suitable contact procedures, interviewer training, follow-up attempts, and reducing unnecessary respondent burden. In some survey settings, weighting or imputation may also be used to handle missing information.

10. Questionnaire Design Principles

A good questionnaire should use clear language, avoid unnecessary ambiguity, follow a logical sequence, and collect information that directly supports the survey objectives. Questions should avoid leading respondents toward a particular answer and should define important terms when respondents may interpret them differently.

A pilot survey or pretest can reveal problems in wording, response options, routing, interview length, and field procedures before full-scale data collection.

11. Data Quality Dimensions

Dimension	Meaning
Accuracy	How closely statistical results represent the phenomena being measured.
Timeliness	How quickly statistics become available after the reference period or event.
Relevance	How well the statistics meet users' needs.
Coherence	How consistently results can be combined with other related statistics.
Accessibility	How easily users can locate, obtain, and understand the statistics.

12. Worked Example: Selecting a Sample

Suppose an agency has a sampling frame containing 10,000 establishments and wants a sample of 500 establishments. Under a simple systematic design, the approximate sampling interval is $k = N/n = 10,000/500 = 20$. After selecting a random starting position between 1 and 20, every 20th establishment is selected.

This example demonstrates why the frame ordering matters: if establishments are ordered in a periodic pattern related to size or activity, a systematic sample may over-represent or under-represent certain groups.

13. Key Takeaways

- Sampling design should follow the objectives, population structure, required precision, and operational constraints.
- Stratification can improve representation and precision when strata are meaningfully different.
- Cluster sampling can reduce operational costs but may increase variance because units within clusters can be similar.
- Sampling error arises from observing a sample rather than the full population; non-sampling errors can occur throughout the survey process.
- Questionnaire pretesting and careful field procedures are important parts of survey quality assurance.
- A competency-aware assessment should test both conceptual knowledge and practical application.

Testing note: This document intentionally contains clear headings, factual definitions, tables, formulas, and examples so that an AI assessment engine can generate a mixture of easy, medium, hard, conceptual, and scenario-based questions.